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PROJECT TASK - 4 DOCUMENTATION REPORT

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7-Segment Display Driver

📌 Abstract

This project implements a 7-Segment Display Driver using Verilog HDL. The circuit converts 4-bit binary input values into corresponding segment control signals required to display decimal digits from 0 to 9 on a 7-segment display.

The design demonstrates combinational logic implementation, binary-to-display conversion, and digital display interfacing. Simulation and waveform verification were performed using EDA Playground and EPWave.

🎯 Objective

The main objectives of this project are:

- To design a 7-Segment Display Driver using Verilog HDL
- To convert binary inputs into segment display outputs
- To understand combinational logic circuits
- To perform simulation and waveform verification
- To display decimal numbers using digital display systems

🛠️ Tools Used

Tool	Purpose
Verilog HDL	Hardware Discription Language
EDA Playground	Online Simulation Platform
Icarus Verilog	Verilog Simulator
EPWave	Waveform Viewer

📖 Theory

7-Segment Display

A 7-segment display is an electronic display device used to display decimal digits. It consists of seven LEDs arranged in the form of segments.

Segment Representation :

```
    a
f       b
    g
 e       c
     d
```

Each segment is controlled individually to display different numbers.

Binary to Segment Conversion

The driver circuit accepts:
4-bit binary input
and generates:
7-bit segment output
which controls the display segments.

Example :

Binary Input	Displayed Number
0000	0
0001	1
0010	2
0011	3

⚙️ Working Principle

- A 4-bit binary number is applied as input.
- The Verilog case statement checks the input value.
- Corresponding segment outputs are generated.
- The required segments turn ON to display the decimal digit.
- Waveform simulation verifies correct segment activation.

🧩 Verilog Module Used

Seven Segment Driver Module

The module:

- Accepts binary input
- Produces segment control output
- Uses combinational logic
- Implements case-based segment mapping

Flies Included

📌 Main Verilog design code
📌 Testbench code
📌 Output screenshots
📌 EPWave waveform screenshots

Simulation Procedure

1. Open EDA Playground
2. Select Verilog language
3. Paste design code into design.sv
4. Paste testbench into testbench.sv
5. Click Run
6. Open EPWave
7. Add signals:

- num
- seg

Output Waveform Analysis

The waveform output shows the relationship between:

- Input binary values
- Segment output patterns

As the input value changes from:

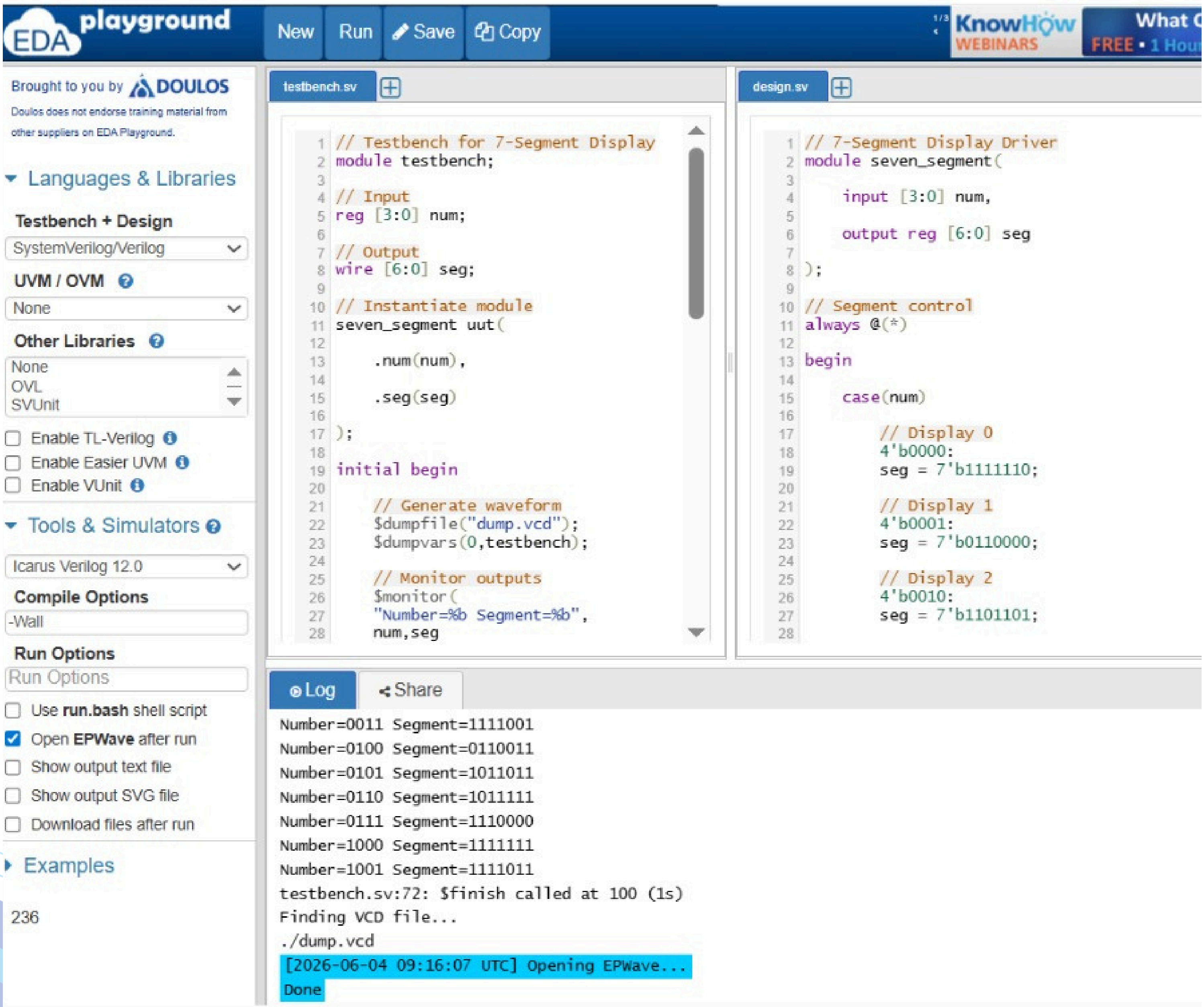
0000 → 1001

the segment outputs change accordingly to display decimal digits from:

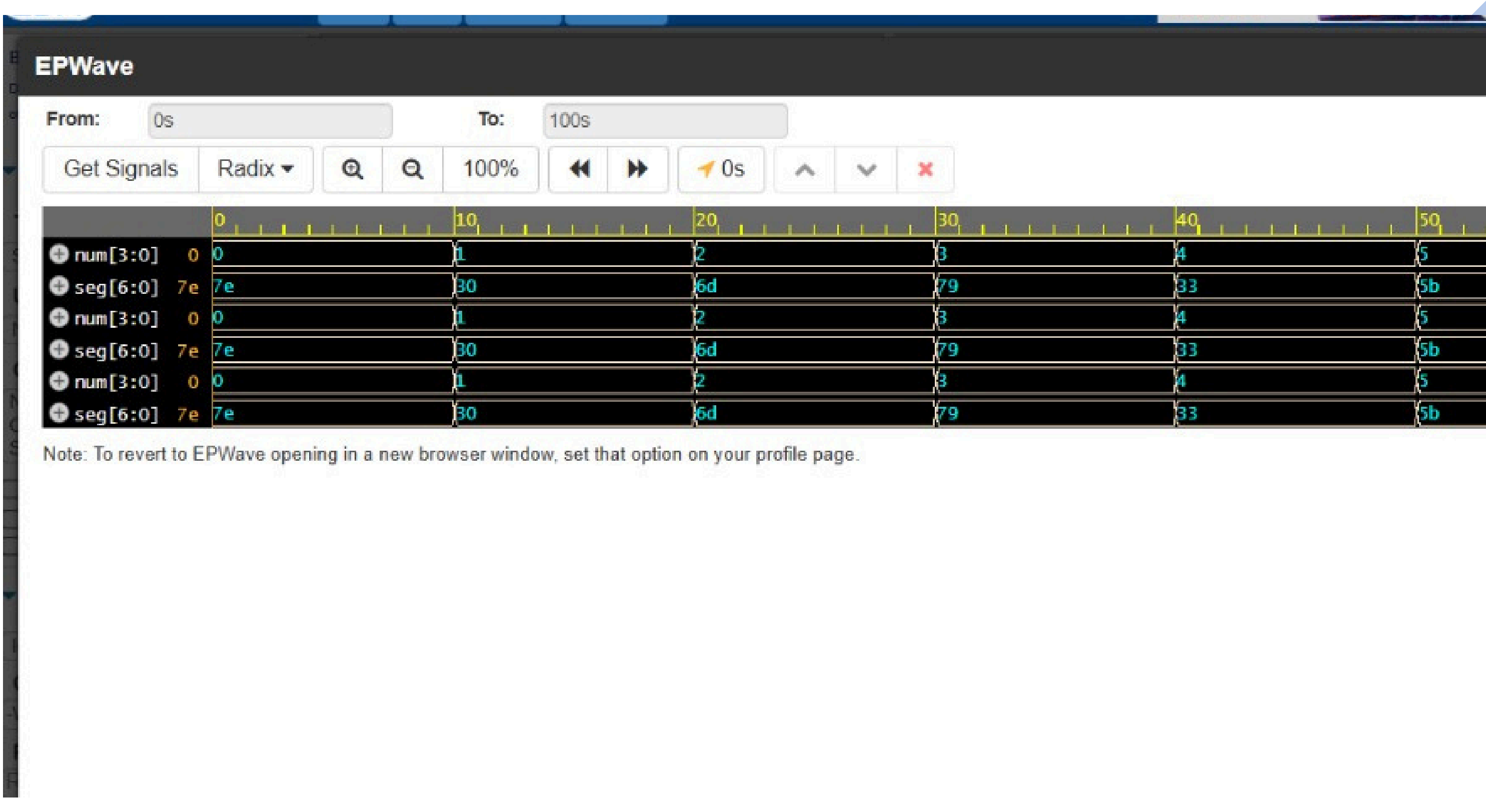
0 → 9

The waveform verifies correct operation of the display driver.

TestBench Code,Main Code



Logic Gates Design-EV Waveforms



References :

- Verilog HDL Documentation
- Digital Electronics Fundamentals
- EDA Playground Simulator Documentation

Conclusion :

The 7-Segment Display Driver was successfully designed and simulated using Verilog HDL. The waveform outputs verified proper conversion of binary input values into segment control signals. The project helped in understanding combinational logic design and digital display interfacing techniques.